

Complete Mathematics Course List

Jiang Yuanqi

Course List (click to jump)

Mathematical Analysis I	<i>undergraduate</i>
Advanced Algebra and Analytic Geometry 2-1	<i>undergraduate</i>
Mathematical Analysis II	<i>undergraduate</i>
Advanced Algebra and Analytic Geometry 2-2	<i>undergraduate</i>
Geometry	<i>upper-level undergraduate</i>
Abstract Algebra	<i>undergraduate</i>
Complex Analysis	<i>upper-level undergraduate</i>
Mathematical Analysis III	<i>undergraduate</i>
Probability Theory	<i>undergraduate</i>
Real Analysis	<i>undergraduate</i>
Combinatorics	<i>upper-level undergraduate</i>
Commutative Algebra	<i>integrated undergraduate-graduate</i>
Lie Algebra	<i>integrated undergraduate-graduate</i>
Analytic Number Theory	<i>integrated undergraduate-graduate</i>
An Introduction to Dynamical System 1	<i>upper-level undergraduate</i>
Functional Analysis	<i>undergraduate</i>
Ordinary Differential Equations	<i>undergraduate</i>
Partial Differential Equations	<i>undergraduate</i>
Numerical Analysis	<i>undergraduate</i>
Lie groups and algebraic groups	<i>integrated undergraduate-graduate</i>
Operations Research	<i>undergraduate</i>
Discrete Structures and their Algorithms	<i>upper-level undergraduate</i>
Representation Theory	<i>special topic</i>
Homological Algebra	<i>special topic</i>
Schubert Calculus	<i>integrated undergraduate-graduate</i>
Geometric Group Theory	<i>integrated undergraduate-graduate</i>

Course Details

Mathematical Analysis I

Instructor: Liu Qing Grade: A

Fall 2023, Nankai University

Main Topics: Real numbers, functions, and basic set-theoretic notions; limits of sequences and functions; continuity; derivatives and differential calculus; applications of derivatives, including mean value theorems, Taylor formula, monotonicity, extrema, convexity, and Newton's method; indefinite integrals; and definite integrals with basic properties and computation.

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Advanced Algebra and Analytic Geometry 2-1

Instructor: Wen Hao Grade: A-

Fall 2023, Nankai University

Main Topics: Polynomials and divisibility, including factorization and symmetric polynomials; determinants and their basic properties; linear systems and elimination; and introductory matrix theory, including matrix operations, rank, inverse matrices, block matrices, and elementary matrices.

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Mathematical Analysis II

Instructor: Liu Qing Grade: A-

Spring 2024, Nankai University

Main Topics: Limits and continuity of multivariable functions; differential calculus of several variables, including partial derivatives, total differentials, directional derivatives, gradients, Taylor expansion, the implicit function theorem, Jacobians, tangent lines and tangent planes, and extrema; integrals involving parameters; multiple integrals; and line and surface integrals.

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Advanced Algebra and Analytic Geometry 2-2

Instructor: Ma Shiguang Grade: A

Spring 2024, Nankai University

Main Topics: Quadratic forms and their classification; linear spaces, subspaces, bases, coordinates, and isomorphisms; linear transformations, eigenvalues and eigenvectors, diagonalization, invariant subspaces, and Jordan theory; lambda-matrices and canonical forms; Euclidean spaces, orthogonality, orthogonal transformations, and least squares; bilinear forms, dual spaces, and symplectic spaces; and basic analytic geometry of lines, planes, and elementary quadratic curves and surfaces.

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Geometry

Instructor: Xu Binbin Grade: A

Spring 2024, Nankai University

Main Topics: Regular polygons and their symmetries; Euclidean geometry of the plane and higher-dimensional spaces; isometries, reflections, rotations, and group actions; finite subgroups of isometry groups and regular polyhedra; affine transformations; projective geometry, including projective spaces, transformations, cross ratios, conics, and duality; and a first introduction to hyperbolic geometry.

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Abstract Algebra

Instructor: Wang Xiuling Grade: A-

Fall 2024, Nankai University

Main Topics: Groups, rings, and fields, including subgroups and quotient groups, normal subgroups, homomorphisms and isomorphisms, cyclic groups, symmetric and alternating groups, ideals and quotient rings, factorization in integral domains, principal ideal domains, Euclidean rings, and splitting fields together with basic questions on algebraic extensions and ruler-and-compass constructions.

Independent study: group actions, Sylow theory, polynomial rings over rings and integral domains, symmetric polynomials, modules, free modules, finitely generated modules over principal ideal domains, finitely generated abelian groups, and standard forms of linear transformations.

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Complex Analysis

Instructor: Gao Yongxin Grade: B+

Fall 2024, Nankai University

Main Topics: Complex numbers and their geometric representation; holomorphic functions, limits, continuity, derivatives, and the Cauchy-Riemann equations; complex integration, the Cauchy theorem, Cauchy integral formula, and consequences such as the maximum principle, Liouville's theorem, and the fundamental theorem of algebra; power series, Taylor and Laurent expansions, isolated singularities, residues, and applications to contour integration; analytic continuation and the Gamma function; and conformal mappings, including linear fractional transformations, the Riemann mapping theorem, and related results.

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Mathematical Analysis III

Instructor: Li Jiaao Grade: C

Fall 2024, Nankai University

Main Topics: Applications of the definite integral in geometry and physics; numerical series, including positive series, alternating series, and infinite products; improper integrals; and function series, including uniform convergence, power series, Taylor series, approximation theorems, and Fourier series.

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Probability Theory

Instructor: Jiang Yiming Grade: A-

Spring 2025, Nankai University

Main Topics: Probability spaces and random events; conditional probability and independence; classical discrete distributions; random variables and distributions; expectation and variance; generating and characteristic functions; multivariate normal distributions; and limit theorems, including the law of large numbers and the central limit theorem.

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Real Analysis

Instructor: Chen Ting Grade: A

Spring 2025, Nankai University

Main Topics: Sets and point-set topology in $\mathbb{R}^n$; Lebesgue outer measure, measurable sets, Borel sets, and regularity; measurable functions and their convergence; the Lebesgue integral and its relation to the Riemann integral; differentiation and indefinite integrals, including bounded variation and absolute continuity.

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Combinatorics

Instructor: Sun Hui and Guo Qianghui Grade: A

Spring 2025, Nankai University

Main Topics: Basic combinatorial language, including sets, relations, functions, posets, elementary counting methods, and combinatorial identities; recurrence relations and generating functions, including ordinary, exponential, and Dirichlet generating functions; inclusion-exclusion and Mobius inversion on posets; special counting sequences and objects such as Catalan numbers, Dyck paths, Stirling numbers, and partitions; and introductory graph theory, including basic notions, trees, and perfect graphs.

Further reading: Stanley, Algebraic Combinatorics: Walks, Trees, Tableaux, and More.

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Commutative Algebra

Instructor: Yu Shizhuo Grade: A

Spring 2025, Nankai University

Main Topics: Group actions and invariant theory; integral domains, ideals, quotient rings, ring homomorphisms, and algebras; Chinese remainder theorem, Dedekind domains, ideal class groups, and Noetherian rings with applications to algebraic number theory; Hilbert's basis theorem and an introduction to algebraic geometry through zero loci and Zariski topology; invariant subrings and the invariant-theoretic form of Hilbert's basis theorem; Grobner bases and Buchberger's algorithm; graded rings and modules; and Noetherian modules, syzygies, and the Hilbert syzygy theorem.

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Lie Algebra

Instructor: Wang Xiuling Grade: A

Spring 2025, Nankai University

Main Topics: Basic notions of Lie algebras, including subalgebras, ideals, quotient algebras, homomorphisms, linear Lie algebras, derivations, direct sums, extensions, and modules; solvable and nilpotent Lie algebras, Jordan-Chevalley decomposition, Engel's theorem, Lie's theorem, the Killing form, and Cartan subalgebras; structure theory of complex semisimple Lie algebras via Cartan subalgebras, root systems, classical root systems, and Dynkin diagrams; classification of complex semisimple Lie algebras; and the first part of representation theory, including complete reducibility and irreducible representations.

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Analytic Number Theory

Instructor: Guo Shaoming, Li Huixi Grade: B+

Fall 2025, Nankai University

Main Topics: Methods and problems in analytic number theory, including the complex-analytic method and the Riemann hypothesis, sieve methods with the Goldbach and twin-prime problems, the circle method and Waring's problem, and exponential sums with applications to Vinogradov's three-primes theorem and lattice-point problems.

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An Introduction to Dynamical System 1

Instructor: Li Ming and Zhou Qi Grade: A

Fall 2025, Nankai University

Main Topics: Discrete dynamical systems, including iterates, invariant sets, topological conjugacy, and hyperbolicity of fixed and periodic points; and symbolic dynamics, including shift spaces, coding of interval maps, kneading sequences, and introductory examples of chaotic behavior such as the Smale horseshoe.

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Functional Analysis

Instructor: Li Lei Grade: A

Fall 2025, Nankai University

Main Topics: Metric and topological spaces; normed linear spaces, including L^p spaces; bounded linear operators and bounded linear functionals, including the Hahn-Banach theorem, open mapping theorem, and Banach-Steinhaus theorem; and Hilbert spaces, orthogonality, and the Riesz representation theorem.

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Ordinary Differential Equations

Instructor: Ma Shiwang Grade: A-

Fall 2025, Nankai University

Main Topics: First-order equations and basic solution methods; existence, uniqueness, and dependence on initial data; higher-order linear equations; linear systems and matrix exponential methods; and nonlinear differential equations, including stability, phase portraits, singular points, and limit cycles.

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Partial Differential Equations

Instructor: Wei Yawei Grade: A-

Fall 2025, Nankai University

Main Topics: Distributions, convolution, and Fourier transform; fundamental solutions; Laplace equation, harmonic functions, and Dirichlet problems; heat equation and wave equation, including initial and boundary value problems.

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Numerical Analysis

Instructor: Wu Chunlin and Zhao Zhiyong Grade: B

Fall 2025, Nankai University

Main Topics: Floating-point computation, error analysis, and stability; interpolation and approximation; numerical integration and differentiation; direct and iterative methods for linear systems; numerical methods for eigenvalue problems; numerical methods for nonlinear equations; and numerical methods for ordinary differential equations.

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Lie groups and algebraic groups

Instructor: Li Ning Grade: A

Spring 2026, Nankai University

Main Topics: A basic introduction to linear algebraic groups, beginning with the algebraic-geometric background of affine and projective varieties, regular maps, and dimension; basic properties of algebraic groups, group actions, and Jordan decomposition; and commutative algebraic groups, including diagonalizable groups, tori, and elementary unipotent groups.

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Operations Research

Instructor: Hu Wei Grade: C+

Spring 2026, Nankai University

Main Topics: Linear programming and simplex methods, including artificial-variable and dual simplex methods; duality and sensitivity analysis; transportation problems; integer and 0-1 programming, including cutting-plane methods; decision analysis; and matrix games.

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Discrete Structures and their Algorithms

Instructor: Guo Long, Sun Hui, and Gu Shanshan Grade: A

Spring 2026, Nankai University

Main Topics: Selected topics in enumerative combinatorics, centered on combinatorial correspondences and generating functions: lattice paths, Catalan and Narayana numbers, reflection arguments, and pattern-avoiding permutations; labeled and plane trees, Prüfer codes, Cayley's formula, and related tree-enumeration identities.

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Representation Theory

Instructor: Yu Jun Grade: 88

Spring 2026, Peking University

Main Topics: Structure and finite-dimensional representation theory of semisimple Lie algebras, including the Killing form, Cartan subalgebras, root systems and Weyl groups; universal enveloping algebras and the PBW theorem; highest-weight theory, Verma modules, complete reducibility, and the Weyl character formula; with selected topics on invariant forms and semisimple Lie groups.

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Homological Algebra

Instructor: Xie Yi Grade: 76

Spring 2026, Peking University

Main Topics: Chain complexes and homology, including singular and cellular homology, long exact sequences, and Mayer-Vietoris; cohomology, the universal coefficient and Kunneth theorems, cup products, and Poincare duality; vector bundles and characteristic classes, including Thom, Euler, and Stiefel-Whitney classes; and an introduction to topological K-theory.

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Schubert Calculus

Instructor: Gao Yibo Grade: 96

Spring 2026, Peking University

Main Topics: Schubert calculus was developed around a central question: how geometric incidence and intersection conditions on Grassmannians and flag varieties can be encoded and computed combinatorially. We first built the framework of root systems, Weyl groups, and Bruhat order, then studied Schubert cells and Schubert varieties and used Plucker coordinates to make the Grassmannian concrete. From this viewpoint, Young tableaux, Schubert polynomials, and pipe dreams became complementary combinatorial models for organizing intersection numbers, multiplication rules, and counting problems. The final part of the course turned to the geometry of Schubert varieties themselves, especially smoothness and singularities, and to the totally nonnegative Grassmannian, where positroids, bounded affine permutations, and Grassmann necklaces describe its cell structure.

Presentation: I gave a presentation on Daoji Huang's paper, Schubert Products for Permutations with Separated Descents.

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Geometric Group Theory

Instructor: Yang Wenyuan Grade: A

Spring 2026, Peking University

Main Topics: An introduction to geometric group theory, starting with free groups, Cayley graphs, Stallings folding, and geometric properties of finitely generated groups such as word metrics, quasi-isometries, group actions, and growth. The course then develops Bass-Serre theory via actions on trees, free products, HNN extensions, and graphs of groups, and continues with hyperbolic geometry and Gromov-hyperbolic spaces, including ideal and Gromov boundaries, quasi-geodesics, the Morse lemma, and basic properties and algorithmic problems of hyperbolic groups.

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